

The Incoherence of Time-Reversal: A Resolution of Loschmidt’s Paradox via the Partition Count, with the Spin Echo as Empirical Witness

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Abstract

Loschmidt’s paradox (1876) asks how the manifest irreversibility of the macroscopic world can be reconciled with the exact time-reversal invariance of the microscopic equations of motion. For a century and a half the standard resolutions—Boltzmann’s statistical improbability, Gibbs’s coarse-graining, the information-theoretic and cosmological-boundary accounts, and environmental decoherence—have shared a single tacit commitment: that time reversal is a well-defined, in-principle-achievable *operation* whose macroscopic consequences merely fail to be observed for contingent reasons. We argue that this commitment is false, that it is empirically testable, and that the experiment which tests it has been performed continuously since 1950 under the name of the *spin echo*.

The argument descends through four depths, each of which resolves the paradox on its own. *First*, we build from first principles an operational account of physical time as a *count of completed oscillatory closures*—the *partition count* M —and prove that this count obeys an exact identity $t = M/f$ with elapsed time, is strictly monotone, and admits no operation that decrements it; the mathematical time-reversal map is an automorphism of the solution set, not a process, and asking whether physical dynamics is “reversible” is a category error. *Second*, we isolate the conflation on which the paradox feeds: it silently identifies three logically independent quantities—the macroscopic observable $\mathcal{O}$, the microstate μ , and the count M —and we prove their genuine distinctness, dissolving the Zermelo recurrence objection at the level of principle. *Third*, we exhibit the empirical witness: the nuclear spin echo recovers the observable by a strictly forward operation on a preserved microstate while the count advances monotonically, and the family of echo experiments—Hahn, Carr–Purcell–Meiboom–Gill, and genuine many-body Hamiltonian reversal—reveals, the more completely one engineers the “reversal,” an irreducible floor set by the system’s own dynamics rather than by engineering imperfection. *Fourth*, we show that real trajectories are sequences of constitutively one-way physical operations, so that the reversibility of the continuous-flow description is a coarse-graining artifact and every purported “rewind” is itself a forward operation that increments the count. The Second Law is thereby recovered not as a statistical tendency but as the operational signature of what time is.

Keywords: Loschmidt’s paradox; irreversibility; arrow of time; partition count; time-reversal invariance; spin echo; Hahn echo; Loschmidt echo; transverse relaxation; second law of thermodynamics; category error.

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1 Introduction

1.1 The paradox

In 1876 Josef Loschmidt raised an objection to Ludwig Boltzmann's program of deriving the second law of thermodynamics from the mechanics of atoms (Loschmidt, 1876, 1877). The objection is disarmingly simple. The microscopic equations of motion governing a system of particles possess a symmetry: take any solution, reverse the sign of time while simultaneously reversing all velocities,

and the result is again a solution. The equations cannot distinguish future from past; they are, in the technical sense made precise in Section 2.2, *time-reversal invariant*.

Yet the macroscopic world is flagrantly asymmetric in time. Heat flows from hot to cold and never the reverse; a drop of ink disperses through water and never reassembles; an egg breaks but is never seen to unbreak (Eddington, 1928). The second law codifies this: the entropy of an isolated system does not decrease. Loschmidt’s challenge was: if the microscopic laws are indifferent to the direction of time, whence the macroscopic arrow? More pointedly—take a gas that has expanded to fill its container, and at some instant reverse every molecular velocity. The reversed state is a perfectly legitimate solution of the same equations, and it will retrace the expansion backwards, the gas re-collecting into the corner it came from, entropy decreasing. The laws permit it. Why is it never seen?

1.2 Why the paradox has resisted resolution

Boltzmann’s reply (Boltzmann, 1877) was statistical: the reversed evolution is not impossible, merely overwhelmingly improbable. The number of microscopic configurations corresponding to a dispersed, high-entropy macrostate vastly exceeds the number corresponding to a concentrated one; an entropy-decreasing evolution must be launched from one of the astronomically rare “conspiratorial” states, and the probability of spontaneously occupying one is of order $e^{-\Delta S/k_B}$, with $\Delta S/k_B$ of order 10^{23} .

This statistical resolution, refined over a century by Gibbs’s coarse-graining (Gibbs, 1902), the information-theoretic reformulation (Jaynes, 1957), the cosmological-boundary account (Penrose, 1989, 2004), the dissipative-structures program (Prigogine, 1980), and environmental decoherence (Zurek, 2003), has become orthodoxy (Table 1). The accounts differ in emphasis but share a single structural commitment, which we call the *reversibility presupposition*:

Time reversal is a well-defined operation. It is in principle achievable; the reversed trajectories exist and are physically realisable; macroscopic irreversibility is the statement that we do not, for contingent reasons (improbability, ignorance, openness, special initial conditions), happen to realise them.

Every standard resolution accepts Loschmidt’s framing and then explains why the reversed evolutions, though available, are not seen. None asks whether the operation Loschmidt invoked—“reverse all the velocities”—is a coherent physical operation at all, or whether, when one attempts to perform it, anything answering to “running the dynamics backwards” actually occurs. That is the question of this paper.

1.3 Thesis and the four depths

Our thesis is that the reversibility presupposition is false, and false in a way that is at once conceptually demonstrable and empirically witnessed. We argue it at four depths, each self-sufficient, each sharper than the last.

1. **Time is a count, and counts cannot decrease** (Section 3). We construct, from scratch, an operational account in which physical time *is* a monotone count of oscillatory closures. “To run time backwards” is to decrement the count; but the enactment of any operation is itself a counting process, hence increments it. The mathematical time-reversal map acts on representations, not on systems; asking whether physical dynamics is “reversible” is a category error.
2. **The conflation the paradox feeds on** (Section 4). Loschmidt’s intuition silently identifies three logically independent quantities—the observable $\mathcal{O}$, the microstate μ , and the count M . We prove their distinctness, which dissolves the Zermelo recurrence objection: a recurrence is a return of μ , not of M , and it is M that constitutes elapsed time.

3. **The empirical witness** (Sections 5–8). The nuclear spin echo is the clean laboratory cousin of Loschmidt’s operation. It recovers the observable by a strictly forward operation on a preserved microstate while the count advances. The hierarchy of echo experiments shows that the better one engineers the reversal, the more sharply an engineering-independent floor appears—the count made visible.
4. **No reverse operation exists** (Section 9). Real trajectories are sequences of constitutively one-way physical operations. The reversibility of the continuous-flow description is a coarse-graining artifact, and every purported “rewind” is a new forward operation incrementing the global count.

Sections 11–14 address objections, state falsifiable predictions, and conclude. Crucially, the resolution is decided not by argument alone but by an experiment that has been running since 1950.

2 Loschmidt’s Paradox in Precise Form

2.1 Hamiltonian dynamics

A classical system of N point particles is specified, at each instant, by the positions $q = (q_1, \dots, q_{3N})$ and momenta $p = (p_1, \dots, p_{3N})$ of all particles; the factor $3N$ counts three spatial directions for each of N particles. The pair (q, p) is a single point in the $6N$ -dimensional *phase space* Γ ; we call it a *microstate*. The dynamics is generated by the *Hamiltonian* $H(q, p)$, equal for our purposes to the total energy, through *Hamilton’s equations*

$$\dot{q}_i = \frac{\partial H}{\partial p_i}, \quad \dot{p}_i = -\frac{\partial H}{\partial q_i}, \quad (1)$$

the overdot denoting the time derivative. Given $(q(0), p(0))$, (1) determines $(q(t), p(t))$ at all later and earlier times. The map sending the initial microstate to the microstate at time t is the *Hamiltonian flow* $\varphi_t : \Gamma \rightarrow \Gamma$; it is deterministic and invertible, with inverse φ_{-t} .

2.2 Time-reversal invariance

Define the *time-reversal map* $R : \Gamma \rightarrow \Gamma$ by $R(q, p) = (q, -p)$: positions unchanged, momenta flipped. For Hamiltonians even in the momenta— $H(q, -p) = H(q, p)$, which holds whenever the kinetic energy is quadratic in p and the forces derive from a position-dependent potential—one verifies from (1) that

$$\varphi_{-t} = R \circ \varphi_t \circ R. \quad (2)$$

Equivalently: if $(q(t), p(t))$ solves Hamilton’s equations, so does $(q(-t), -p(-t))$. This is what it means for the dynamics to be *time-reversal invariant*: the set of solutions is mapped onto itself by the combined operation of reversing time and flipping momenta. It is a property of the *solution set*.

Remark 2.1 (The hinge of the whole problem). Equation (2) is a statement about *mathematical solutions*. It says a certain transformation carries one solution to another. It does not say—and this distinction is the hinge on which the entire paradox turns—that there exists a *physical operation*, performable in a laboratory, whose execution carries a physical system from the first solution to the second. We will return to this in Sections 3 and 9.

2.3 Liouville and Poincaré

Two classical results sharpen the paradox. *Liouville’s theorem* (Liouville, 1838): the flow preserves phase-space volume, $\text{vol}(\varphi_t(A)) = \text{vol}(A)$ for every region $A \subseteq \Gamma$; the flow stirs phase space

Table 1: Standard resolutions of Loschmidt’s paradox. Each accepts that time reversal is a coherent, in-principle-achievable operation and explains only why its consequences are not observed (third column).

Resolution	Account of why reversal is not seen	Residual commitment
Boltzmann statistical (Boltzmann, 1877)	Reversed trajectories require astronomically improbable microstates ($P \sim e^{-\Delta S/k_B}$).	Reversal is possible, merely improbable.
Gibbs coarse-graining (Gibbs, 1902)	Entropy increases only for a smeared distribution; fine-grained entropy is constant by Liouville.	Irreversibility is observer-relative.
Information-theoretic (Jaynes, 1957)	Entropy measures the observer’s ignorance of the microstate.	Irreversibility is epistemic; reversal remains coherent.
Cosmological boundary (Penrose, 1989, 2004)	The arrow stems from a low-entropy initial state of the universe.	Explains the gradient, not the absence of local reversal.
Decoherence (Zurek, 2003)	Environmental entanglement destroys the phase relations reversal needs.	The closed-system reversal is granted as coherent.

without compressing it, so irreversibility cannot be a contraction of volume. *Poincaré’s recurrence theorem* (Poincaré, 1890): if the accessible phase space has finite volume (as for a system of bounded energy in a bounded region), almost every microstate returns arbitrarily close to its initial value, infinitely often. Ernst Zermelo (Zermelo, 1896) wielded this against Boltzmann: if every state recurs, no quantity can increase monotonically forever. Boltzmann’s reply (Boltzmann, 1896) was again statistical—the recurrence times are unimaginably vast. We record recurrence here because, as Section 4 shows, the partition account dissolves Zermelo’s objection at a different level: a recurrence of the *microstate* is not a recurrence of the *count*, and it is the count that constitutes elapsed time.

2.4 The objection, sharply

Loschmidt’s objection, in its sharpest form: a gas prepared at $t = 0$ in a low-entropy macrostate (all molecules in one corner) evolves to time T , filling its container (high entropy). Apply R at T . By (2), the subsequent forward evolution carries the system back through its expansion in reverse order until, at $2T$, it sits again in the corner; entropy will have decreased. The trajectory is a bona fide solution of the same equations. *The laws sanction this; the second law forbids it; how can both be true?*

2.5 The standard resolutions and their common structure

Each entry of Table 1 is correct within its domain. What they share is the residual commitment in the third column: the *operation* of time reversal is treated as meaningful and realisable, and only its *consequences* are explained away. The present paper attacks the commitment itself.

3 Time as a Partition Count

We now build, from first principles and without reference to any prior framework, an operational account of what physical time *is*, in which the asymmetry we seek is not imposed but constitutive.

3.1 Time is read from a counter

Ask how elapsed time is established in any actual physical measurement. The answer is always the same: one counts cycles of a reference oscillator. A pendulum clock counts swings; a quartz watch counts crystal vibrations; the SI second counts a fixed number—9,192,631,770—of oscillations of the radiation associated with a hyperfine transition of caesium-133 (Essen and Parry, 1955). In every case the procedure is identical: fix a reference oscillator, count its completed cycles, report the count (suitably scaled) as elapsed time.

We take this literally. Time, as an operationally defined physical quantity, *is* a count of completed oscillatory cycles.

Definition 3.1 (Oscillatory closure). Let a system possess a degree of freedom describable by an angle θ on the circle $[0, 2\pi)$. An *oscillatory closure* is the event of θ completing one full traversal of $[0, 2\pi)$ and returning to its starting value. A closure is a purely *topological* event—the completion of a loop in configuration space—and its occurrence makes no reference to any unit of duration.

Definition 3.2 (Partition count). The *partition count* $M(t) \in \mathbb{Z}^+$ of a system relative to a reference oscillator is the integer number of oscillatory closures of that oscillator completed up to time t . The term “partition” records that each closure partitions the system’s history into a before and an after: the closure is the irreducible unit by which one stretch of existence is marked off from the next.

Remark 3.3 (Non-circularity). One might object that a “cycle” already presupposes a duration, making the account circular. It does not. A closure is defined by the angular condition “ θ has traversed $[0, 2\pi)$ and returned,” a statement about the *geometry* of the trajectory in configuration space, not about how long the traversal took. Two oscillators are compared by the dimensionless ratio of their counts, M_A/M_B . One is promoted to the standard, and a unit of duration is *defined* by stipulating how many of its closures constitute that unit (for the SI second, 9,192,631,770 closures of the caesium standard). Frequency in hertz is then the ratio of a system’s count to the caesium count. Duration and frequency are *derived* from counting; counting does not presuppose them. This is the conceptual reverse of the usual order, in which frequency (= cycles per second) presupposes the second; here the count is primitive and the second is the derived stipulation.

3.2 Bounded systems oscillate

The partition count is well defined whenever there is an oscillator to count, and bounded physical systems always supply one.

Proposition 3.4 (Oscillatory necessity). *A mechanical system confined to a region of finite spatial extent and possessing finite energy executes bounded motion; on any such bounded energy surface no coordinate can escape monotonically to infinity, and (in the absence of fixed points, excluded for systems at positive temperature) the motion is oscillatory: every trajectory returns repeatedly to the neighbourhood of its prior configurations.*

Sketch. Confinement to a bounded region with finite energy makes the accessible phase space bounded, hence (being closed) compact and of finite volume. A coordinate escaping monotonically would leave every bounded region, contradicting confinement. By Poincaré recurrence (Section 2.3) almost every trajectory returns arbitrarily near its initial condition; with no fixed points available, the recurrent returns are traversed cyclically, i.e. oscillatorily. $\square$

Proposition 3.4 guarantees the count of Definition 3.2 for any bounded system, including the trapped nuclear spins of Section 5.

3.3 The Time–Count Identity

Theorem 3.5 (Time–Count Identity). *For a reference oscillator of frequency f (defined, per Remark 3.3, as the number of its closures per unit of the standard), the partition count and the elapsed time satisfy*

$$M(t) = f t, \quad \text{equivalently} \quad t = \frac{M}{f}, \quad (3)$$

exactly, and the rate of counting equals the frequency, $dM/dt = f$.

Proof. By definition of frequency the oscillator completes f closures per unit of standard time; in elapsed time t it completes ft closures, which is the count by Definition 3.2. The relation $dM/dt = f$ is its differential form. The continuous-time parameter $t \in \mathbb{R}$ of the formalism is the $f \rightarrow \infty$ idealisation of the integer count, in which the discrete steps become unresolvably fine. $\square$

Read from right to left, (3) would be circular (f presupposes t); read as the definition of the derived continuous parameter t in terms of the primitive count M , it is not. The identity records that the count *is* the time, in units fixed by the choice of standard.

3.4 Monotonicity and the incoherence of decrement

Theorem 3.6 (Strict monotonicity). *The partition count is strictly increasing along any physical evolution: if $t_2 > t_1$ and at least one closure occurs in $[t_1, t_2]$, then $M(t_2) > M(t_1)$.*

Proof. By Theorem 3.5, $M(t_2) - M(t_1) = f(t_2 - t_1) > 0$. $\square$

Theorem 3.7 (Incoherence of decrement). *No physical operation can decrease the partition count. In particular, the mathematical time-reversal map R of Section 2.2 has no realisation as a count-decrementing operation.*

Proof. Suppose an operation $\mathcal{A}^*$ purported to take the count from M to $M - k$ with $k > 0$. Its execution is itself a physical process occupying a positive interval $\Delta t > 0$, during which, by Theorem 3.6, the reference oscillator completes at least $f \Delta t > 0$ closures. The count after the operation therefore exceeds the count before it by at least $f \Delta t$, contradicting the supposition that it decreased. The map R acts on the solution set of the equations (Section 2.2); applying it is an act of mathematical substitution, not a laboratory procedure, and it does not act on M at all. There is no operation whose execution decrements the count, because executing anything increments it. $\square$

Theorem 3.8 (Inverses do not decrement the count). *For any operation $\mathcal{A}$ whose mathematical inverse $\mathcal{A}^{-1}$ exists as a physical operation, the application of $\mathcal{A}^{-1}$ is itself an operation that increments the partition count. Hence $\mathcal{A}^{-1}$ does not undo $\mathcal{A}$ at the count level: the count after $\mathcal{A}^{-1} \circ \mathcal{A}$ strictly exceeds the count before $\mathcal{A}$.*

Proof. The application of $\mathcal{A}^{-1}$ consumes time $\Delta t > 0$, during which the reference oscillator generates at least $f \Delta t \geq 1$ closures. The phase-space coordinates may return to their values before $\mathcal{A}$; the count strictly increases. $\square$

Theorem 3.7 is the partition-level statement of the arrow of time: not that decrement is improbable, but that the very attempt to perform a decrementing operation is a counting process, hence incrementing. Theorem 3.8 catches the right nuance—it does not deny that some operations have mathematical inverses; it observes that even when they do, applying the inverse is a forward partition event. *The phase-space trajectory may close on itself; the count cannot.*

Corollary 3.9 (The category error). *The question “is time-reversal possible in physical systems?” is a category error. Time-reversal invariance (2) is a relation on solution representations; physical irreversibility is a property of processes. The first is a feature of how we write the dynamics; the second is a feature of what the dynamics does. Loschmidt’s paradox arises by treating a relation on representations as though it were an operation on systems, then asking why its consequences are not observed.*

The second law, in this reading, is not a statistical tendency layered on reversible foundations; it is the observation that the count cannot decrease because counting is what time is.

4 The Three-Quantity Separation

The conceptual core of the paper is the recognition that Loschmidt’s intuition—“if the trajectory can be retraced, the system returns to its past”—silently identifies three quantities that must be kept distinct. We name them and prove their independence; the spin echo of Section 5 will make the distinctions visible and measurable.

Definition 4.1 (The three quantities). For a system of many constituents we distinguish:

- (i) the *microstate* $\mu(t)$: the complete instantaneous specification of every constituent (for the gas, the full point of phase space; for the spins of Section 5, the full list of individual spin phases);
- (ii) the *observable* $\mathcal{O}(t)$: a coarse, macroscopically accessible function of the microstate (for the gas, the spatial density profile; for the spins, the net transverse magnetisation), in general *many-to-one*, so that many microstates share one observable value;
- (iii) the *partition count* $M(t)$: the monotone count of Section 3, recording how much has happened.

Principle 4.2 (The conflation). Loschmidt’s reversibility intuition rests on the implicit identification $\mathcal{O} \equiv \mu \equiv M$: that recovering the observable ($\mathcal{O}(2T) = \mathcal{O}(0)$) is the same as restoring the microstate ($\mu(2T) = \mu(0)$), which is the same as returning to the past ($M(2T) = M(0)$). The three are distinct, and conflating them is what makes “the trajectory returned” sound like “the system went back.”

Theorem 4.3 (Genuine distinctness). *The three quantities of Definition 4.1 are logically independent:*

- (i) $\mathcal{O}$ may return to a past value while μ does not: many microstates share an observable, so $\mathcal{O}(t_2) = \mathcal{O}(t_1)$ does not imply $\mu(t_2) = \mu(t_1)$;
- (ii) μ may return to a past value while M does not: a Poincaré recurrence restores the microstate but, by Theorem 3.6, the count has strictly increased;
- (iii) M never returns to a past value at all (Theorem 3.7).

Proof. (i) is the many-to-one character of $\mathcal{O}$ in Definition 4.1(ii). (ii): at a recurrence time t_2 with $\mu(t_2) \approx \mu(t_1)$, Theorem 3.6 gives $M(t_2) > M(t_1)$ because closures accumulated in the interim. (iii) is Theorem 3.7. $\square$

Corollary 4.4 (Dissolution of the Zermelo objection). *Zermelo’s recurrence objection (Section 2.3) is dissolved at the level of principle. A Poincaré recurrence is a return of μ , not of M ; since elapsed time is constituted by M (Theorem 3.5), the recurrence does not return the system to an earlier time, and no monotone temporal quantity is threatened by it. The recurrence theorem is true and irrelevant to the arrow.*

4.1 Recovering the two-entropy picture

The separation refines the familiar entropy bookkeeping. The Boltzmann entropy $S_{\text{kin}} = k_B \ln \Omega$, with Ω the number of microstates compatible with a macrostate, is a function of $\mathcal{O}$ and μ and is formally reversible under velocity inversion: the mathematical operation R carries $S_{\text{kin}}(\mu_f) \mapsto S_{\text{kin}}(\mu_0)$. But there is a second, *categorical* contribution that tracks the number of committed distinctions—the count of partition events, $S_{\text{cat}} = k_B M \ln n$ for M distinctions each among n alternatives—and this is a function of M alone. Velocity inversion does not touch M (Theorem 3.7), so S_{cat} is unaffected, and the total $S = S_{\text{kin}} + S_{\text{cat}}$ cannot be returned to its initial value even granting perfect kinetic reversal. The “entropy that Loschmidt could in principle reverse” is exactly the μ -dependent part; the “entropy that he cannot” is exactly the count. The two-entropy split is the Three-Quantity Separation seen thermodynamically.

Theorem 4.3 already dissolves the paradox at the level of principle. But a principle is not an experiment. The remainder of the paper shows that the spin echo realises this separation in the laboratory—and, in the Loschmidt-echo progression, measures the separating quantity directly.

5 The Nuclear Spin Echo: A Complete Account

There is a tradition of treating Loschmidt’s velocity reversal as a thought experiment forever beyond the laboratory, since one cannot grab 10^{23} molecules and flip their momenta. This is a mistake. A close and informative laboratory cousin of Loschmidt’s operation has been performed routinely since 1950 under the name of the *spin echo* (Hahn, 1950). This section is a self-contained primer; a reader who knows magnetic resonance may skim it.

5.1 Spins, magnetic moments, Larmor precession

Many atomic nuclei possess an intrinsic angular momentum, *spin*, and with it a *magnetic moment* μ : the nucleus behaves like a tiny bar magnet. Placed in a strong uniform static field $\mathbf{B}_0$ along the laboratory z -axis, a moment experiences a torque that makes it *precess*—sweep a cone about the field, like a leaning spinning top—at the *Larmor frequency*

$$\omega_0 = \gamma B_0, \tag{4}$$

where γ is the gyromagnetic ratio. For an ensemble of $\sim 10^{22}$ identical nuclei the individual moments sum to a macroscopic *magnetisation* $\mathbf{M}$. In equilibrium $\mathbf{M}$ lies along $\mathbf{B}_0$; the transverse components cancel because the individual phases are uniformly distributed around the cone.

5.2 The rotating frame

It is convenient to describe the motion in a frame rotating about z at ω_0 . In this *rotating frame* a moment precessing exactly at ω_0 appears stationary; one precessing at $\omega_0 + \Delta\omega$ appears to rotate slowly at the *offset* $\Delta\omega$. Throughout, “phase” φ means the angle of a moment’s transverse component in the rotating frame.

5.3 Radio-frequency pulses

A short burst of oscillating field perpendicular to $\mathbf{B}_0$ and tuned to ω_0 rotates $\mathbf{M}$ by a controllable angle. Two pulses concern us: the 90° *pulse*, which tips equilibrium magnetisation into the transverse (xy) plane, creating a coherent transverse magnetisation with all moments initially in phase; and the 180° (*refocusing*) *pulse*, which rotates the magnetisation half a turn about a transverse axis, with the effect—central to the echo—of replacing each phase φ by $-\varphi$.

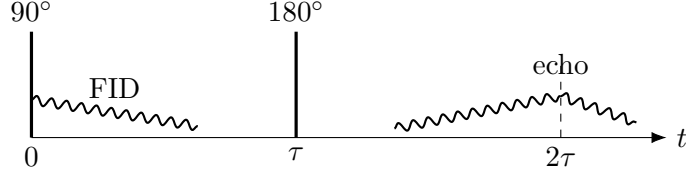


Figure 1: The Hahn spin-echo sequence. A 90° pulse at $t = 0$ initiates the FID; a 180° refocusing pulse at $t = \tau$ inverts the accumulated phases; the signal rephases into an echo at $t = 2\tau$. The echo is reduced relative to the initial FID by the irreversible factor $e^{-2\tau/T_2}$.

5.4 Free induction decay and the two dephasings

Apply a 90° pulse at $t = 0$: a maximal transverse signal appears (a precessing magnetisation induces a voltage in a pickup coil) and immediately begins to decay—the *free induction decay* (FID)—because the moments do not all precess at exactly the same frequency and so *dephase*, fanning out until their vector sum shrinks. Two physically distinct causes of dephasing must be told apart; the whole argument hinges on the distinction.

Inhomogeneous dephasing (T_2^*). The static field is never perfectly uniform, and different nuclei sit in slightly different environments. A nucleus where the field is $B_0 + \delta B_i$ precesses at a *static* offset $\Delta\omega_i = \gamma \delta B_i$ fixed for the experiment, accumulating the exactly determined phase

$$\varphi_i^{\text{stat}}(t) = \Delta\omega_i t. \quad (5)$$

The spread of fixed offsets makes the signal decay on a timescale T_2^* . This dephasing is *deterministic* and, as we will see, *reversible*.

Homogeneous dephasing (T_2). Superimposed are *fluctuating* local fields—each nucleus is jostled by the time-varying fields of its neighbours, by molecular tumbling, by diffusion through gradients (Bloch, 1946; Bloembergen et al., 1948). The phase accumulated from this randomly varying part,

$$\varphi_i^{\text{fluc}}(t) = \int_0^t \delta\omega_i(t') dt', \quad (6)$$

is stochastic, with some short correlation time. This dephasing proceeds on the *true* or *homogeneous* transverse relaxation time T_2 . It is *stochastic* and, as we will see, *irreversible*. The two timescales combine for the observable envelope as $1/T_2^* = 1/T_2 + 1/T_2^{\text{inh}}$, with $T_2^* \leq T_2$; the raw FID decays faster than the true relaxation would dictate. The spin echo strips away the inhomogeneous part and reveals T_2 .

5.5 The Hahn echo

Hahn’s 1950 sequence (Hahn, 1950), sketched in Figure 1:

1. at $t = 0$, a 90° pulse creates coherent transverse magnetisation; the ensemble dephases;
2. at $t = \tau$, a 180° pulse replaces each $\varphi_i(\tau)$ by $-\varphi_i(\tau)$;
3. the ensemble precesses on under the *same* field;
4. at $t = 2\tau$, the signal spontaneously reappears as an *echo*.

The mechanism is exact for the static part. After the 180° pulse, nucleus i carries phase $-\Delta\omega_i\tau$ and continues to accumulate at the same rate, so at 2τ its static phase is

$$-\Delta\omega_i\tau + \Delta\omega_i(2\tau - \tau) = -\Delta\omega_i\tau + \Delta\omega_i\tau = 0. \quad (7)$$

The cancellation is *independent of* $\Delta\omega_i$: every nucleus, whatever its fixed offset, returns to zero phase at 2τ simultaneously. The fan closes; the moments realign; the macroscopic transverse magnetisation—the observable—reappears.

5.6 The echo deficit

The echo is not as tall as the original FID. The fluctuating part (6) is not cancelled by the pulse, because the random phase a nucleus accumulates between τ and 2τ is statistically independent of what it accumulated between 0 and τ ; inverting the past phase cannot pre-compensate independent future kicks. The residual random phase degrades the realignment, and the echo amplitude is reduced by

$$\frac{A(2\tau)}{A(0)} = e^{-2\tau/T_2}, \quad (8)$$

governed by the *true* relaxation time T_2 , not by T_2^* . Measuring echo height versus τ isolates T_2 : the echo subtracts away the reversible inhomogeneous dephasing and lays bare the irreversible homogeneous part.

6 The Echo Recovers Configuration, Not History

We now formalise the spin echo in the language of Section 4. The microstate of the spin ensemble in the rotating frame is the full list of phases $\mu(t) = (\varphi_1(t), \dots, \varphi_N(t))$; the observable is the normalised transverse magnetisation

$$\mathcal{O}(t) = \frac{1}{N} \left| \sum_{i=1}^N e^{i\varphi_i(t)} \right|; \quad (9)$$

and the count is $M(t) = ft$ for the spectrometer’s reference oscillator (Theorem 3.5).

Theorem 6.1 (Echo-Configuration Recovery). *In the Hahn-echo sequence, restricted to the inhomogeneous (T_2^*) dephasing:*

- (i) *the observable returns, $\mathcal{O}(2\tau) = \mathcal{O}(0)$, by the $\Delta\omega_i$ -independent refocusing (7);*
- (ii) *the microstate is never lost: at every instant $t \in [0, 2\tau]$ each $\varphi_i(t)$ is an exactly determined function of the fixed offset $\Delta\omega_i$ and the pulse history, so $\mu(t)$ carries the complete information of $\mu(0)$ forward without erasure;*
- (iii) *the count increases monotonically, $M(2\tau) > M(\tau) > M(0)$, by Theorem 3.6.*

Consequently the echo is the construction of a new microstate $\mu(2\tau)$, reached by strictly forward operations, whose observable happens to coincide with a past observable. No past microstate is re-occupied and no past count is restored.

Proof. (i) is Equation (7) summed over the ensemble in (9). (ii): under inhomogeneous dephasing the phase is (5) before the pulse and its sign-flipped continuation after; both are deterministic functions of the fixed $\Delta\omega_i$, a permanent property of nucleus i carried in the microstate, so no φ_i is ever undetermined or erased. (iii) is Theorem 3.6 over $[0, 2\tau]$. The final clause follows: $\mathcal{O}(2\tau) = \mathcal{O}(0)$ by (i) while $M(2\tau) > M(0)$ by (iii), so by Theorem 4.3 the coincidence of observables is neither a recurrence of microstate nor of count. $\square$

Remark 6.2 (The decay of the FID is not a loss). Theorem 6.1(ii) makes precise a point easily missed: the inhomogeneous decay of the FID is a shrinking of the *observable* $\mathcal{O}$, not a loss of information from the *microstate* μ . The moments fan out, the vector sum cancels, the signal vanishes—yet every phase remains exactly where the deterministic dynamics put it. This is the many-to-one gap of Definition 4.1(ii): a small observable is compatible with a fully determinate, information-rich microstate. The echo exploits this gap. It does not retrieve lost information; it re-aligns information that was never lost.

Theorem 6.1 is the sharpest illustration of why “the signal came back” does not mean “the system went back.” In the spin echo one can point to all three quantities at once: the observable that returned, the microstate that went forward intact, and the count that advanced and cannot return. The naive reading collapses them; the experiment pulls them apart.

7 The Refocusability–Partition Exclusion Theorem

Theorem 6.1 was restricted to the inhomogeneous part. We now explain *why* the refocusing pulse recovers the static dephasing but is powerless against the fluctuating dephasing. The answer is a structural exclusion, and it identifies the irreversible part of relaxation with the count-incrementing part.

Definition 7.1 (Partition event). A *partition event* for a constituent is an interaction with other degrees of freedom (other spins, the lattice, the diffusive environment) that transfers which-constituent information into those degrees of freedom and commits it there: after the event, recovering the constituent’s pre-event phase would require reading the correlated state of the environment, not the constituent alone. Each fluctuating kick $\delta\omega_i(t') dt'$ in (6) is the phase imprint of such an event.

Definition 7.2 (Refocusable contribution). A dephasing contribution is *refocusable* at τ if there exists an operation P applied at τ (acting on the present microstate $\mu(\tau)$) such that the subsequent forward evolution restores the observable, $\mathcal{O}(2\tau) = \mathcal{O}(0)$, with respect to that contribution.

Theorem 7.3 (Refocusability–Partition Exclusion). *A dephasing contribution is refocusable at τ if and only if it created no partition event in $[0, \tau]$. Equivalently: a contribution is refocusable iff the phase it produces at 2τ is a determinate function of the present microstate $\mu(\tau)$; a contribution that incremented the partition count through committed interactions is, by construction, not such a function, and is therefore not refocusable.*

Proof. ($\Leftarrow$) Suppose the contribution created no partition event. Then no which-spin information was committed to the environment, and the contribution is a deterministic function of properties carried in the microstate—for the static offsets, the fixed $\Delta\omega_i$. The future phase increment over $[\tau, 2\tau]$ equals the past increment over $[0, \tau]$, because the generating offset is unchanged. The operation $P = “\varphi_i \mapsto -\varphi_i”$ (the 180° pulse) then yields the cancellation (7), restoring $\mathcal{O}$.

($\Rightarrow$) Contrapositive: suppose the contribution created partition events. By Definition 7.1 the phase increment over $[\tau, 2\tau]$ is generated by interactions whose realised values are committed to the environment and are statistically independent of the increment over $[0, \tau]$; knowing $\mu(\tau)$ does not determine the future increment. Any operation P at τ acts only on $\mu(\tau)$; it can negate the accumulated past phase but cannot pre-compensate an increment that is not a function of $\mu(\tau)$. Hence the residual phases at 2τ have non-zero variance, the vector sum (9) is degraded, and $\mathcal{O}(2\tau) < \mathcal{O}(0)$. $\square$

Corollary 7.4 (The echo deficit is the count made visible). *The echo-amplitude deficit (8), $1 - e^{-2\tau/T_2}$, is a calibrated, directly observed measurement of the accumulated partition events in $[0, 2\tau]$. The reversible part of relaxation is exactly the configurational (μ -determined) part; the irreversible part is exactly the count-incrementing part. The homogeneous relaxation envelope is the count M rendered as a laboratory signal.*

This is the structural lock the whole construction needed: the boundary between what an idealised reversal can and cannot recover coincides *exactly* with the boundary between the configurational quantities $(\mathcal{O}, \mu)$ and the count M . The experiment does not merely illustrate the Three-Quantity Separation; it measures the separating quantity.

8 The Loschmidt Echo Progression and the Irreducible Floor

The Hahn echo reverses only the inhomogeneous, single-particle dephasing. A natural research program, pursued for seven decades, asks: how much more of the dynamics can be reversed if we engineer the refocusing more ambitiously? The answer is the decisive empirical structure of this paper.

Hahn echo (Hahn, 1950). A single 180° pulse reverses static single-spin dephasing. Residual: $e^{-2\tau/T_2}$, set by the fluctuating environment.

Carr–Purcell–Meiboom–Gill (CPMG) (Carr and Purcell, 1954; Meiboom and Gill, 1958). A *train* of refocusing pulses, spaced more finely than the correlation time of the fluctuations, suppresses part of the diffusive and slowly-fluctuating dephasing as well. The faster the pulsing, the more is recovered—but the recovery saturates: a residual decay survives, governed by the genuinely fast, irreducible fluctuations.

Magic echo and many-body reversal (Rhim et al., 1971). By engineering the radio-frequency drive one can effectively invert the sign of the *dipolar coupling Hamiltonian* itself, $H \rightarrow -H$, reversing not just single-spin precession but the many-body interacting evolution. This is the closest physical approximation to Loschmidt’s “reverse all velocities” ever realised.

Polarization / Loschmidt echo (Gorin et al., 2006; Jalabert and Pastawski, 2001; Levstein et al., 2001). The most refined many-body reversal experiments measure the *fidelity* with which the reversed evolution reconstructs the initial state. As the engineered reversal is made ever more perfect, the reconstruction fidelity does not approach unity. It approaches a floor whose decay rate becomes *independent of the perturbation strength*—set by the system’s own internal dynamics rather than by the residual imperfection of the engineering (Gorin et al., 2006; Jalabert and Pastawski, 2001).

Principle 8.1 (The irreducible floor). Across the progression, improved engineering of the reversal exposes an engineering-independent floor. The better one approximates Loschmidt’s operation, the more sharply one isolates a residual irreversibility that does not diminish with further engineering, because it is not an artifact of imperfect control but the signature of accumulated partition events.

Theorem 8.2 (Floor unification). *The irreducible floor of the Loschmidt-echo progression is a single object viewed three ways: (a) the homogeneous-relaxation limit $e^{-2\tau/T_2}$ of the configurational recovery (Corollary 7.4); (b) the non-decrementability of the partition count (Theorem 3.7); (c) the positivity of the smallest committed correlation that any refocusing operation, acting only on the present microstate, cannot pre-compensate (Theorem 7.3).*

Proof. (a) $\Leftrightarrow$ (c): by Theorem 7.3 the non-refocusable part is precisely the partition-event part, whose envelope is (8); the floor is that envelope’s non-vanishing as engineering improves, because partition events, once committed, are not functions of the present microstate. (c) $\Leftrightarrow$ (b): a partition event is, by Definition 7.1, a committed increment of M ; by Theorem 3.7 no operation decrements M , so no refocusing can annul a committed event. The three descriptions therefore have the same content: a positive floor exists iff at least one partition event has been committed, and the count records exactly those. $\square$

The lab has been running Loschmidt’s own experiment for seventy years. Every refinement confirms the floor. *The experiment named after Loschmidt refutes Loschmidt’s premise:* the more completely one engineers the reversal, the more cleanly one measures the quantity that cannot be reversed.

9 No Reverse Operation Exists

The arguments so far live within the framework of dynamical systems—state spaces, flows, trajectories as curves. We now move to a deeper level. Real physical processes are not curves in continuous phase space; they are sequences of physical *operations*, each constitutively one-way.

Definition 9.1 (Operation; constitutively one-way). A physical *operation* $\mathcal{A} : \Gamma \rightarrow \Gamma$ is a partial map between observable states implemented by a specific physical mechanism; different mechanisms producing the same input–output map are different operations. An operation $\mathcal{A}$ is *constitutively one-way* if no operation $\mathcal{A}^*$ exists in physical reality such that $\mathcal{A}^* \circ \mathcal{A} = \text{id}$ as physical processes (not merely as mathematical maps).

The empirical world is full of constitutively one-way operations. Fertilisation has no mechanism called “un-fertilisation”; combustion has no single reverse operation (reduction, photosynthesis, and electrolysis are different operations with different intermediates); mitosis is not run backwards by cell fusion (a different mechanism); crystallisation is not dissolution-in-reverse. Consider a maize seed in your hand. “How was this produced?” is a coherent question with a determinate answer: pollination, fertilisation, zygote and endosperm formation, mitotic division, differentiation, starch accumulation, dehydration, dormancy. “Can this be reached by reversing time?” is incoherent: each constituent operation has no reverse mechanism, only different forward operations whose mathematical action happens to match an inverse. The seed is the end-product of a sequence of constitutively one-way operations; its existence in your hand is testimony to their directionality.

Theorem 9.2 (Constitutive Asymmetry). *Every physical trajectory is a sequence of operations $\{\mathcal{A}_1, \dots, \mathcal{A}_k\}$, each constitutively one-way. The composite has no reverse trajectory in the operational sense: no sequence of operations exists whose enactment undoes the forward trajectory as a physical process.*

Sketch. Suppose a reverse trajectory $\tilde{\mathcal{A}}_1 \circ \dots \circ \tilde{\mathcal{A}}_k$ existed, composing to the identity-as-a-process. Then each $\tilde{\mathcal{A}}_i$ inverts $\mathcal{A}_i$ as a process. Three cases. If $\tilde{\mathcal{A}}_i$ does not exist, the reverse is unavailable. If it exists but is a different operation, it has different mechanism and intermediates, so $\tilde{\mathcal{A}}_i \circ \mathcal{A}_i$ is a sequence of two distinct operations, not the identity-as-a-process. If it exists and is the same operation, $\mathcal{A}_i$ is its own inverse—possible only for symmetry transformations that do not advance the partition structure, which generic operations (ionisation, detection, fertilisation) are not. In all generic cases the reverse trajectory does not exist as a physical process. $\square$

Corollary 9.3 (Reversibility is a coarse-graining artifact). *The time-reversibility of the continuous-flow description $\dot{x} = f(x, t)$ is an artifact of replacing operation-sequences with continuous flows. The flow averages over the operational structure, and the averaging happens to be invariant under $t \rightarrow -t$. The averaging is not itself a physical process.*

Principle 9.4 (Rewind-as-Forward). A “rewound” sequence is not the original sequence run backward. It is a new forward sequence whose state-content is the reverse-ordered list of states from the original. The rewinding itself is a forward operation that increments the global partition count.

A film projector showing a movie in reverse runs forward in time as it does so: photons are emitted forward, the audience perceives them forward, the operator’s watch ticks forward. The frames appear in reverse order, but the reverse-ordering is a forward physical process. The same holds for every claimed time-reversal in physics. A Maxwell demon reversing molecular velocities must act forward, measure forward, expend forward thermodynamic resources; the “reversed” trajectory that ensues is a new forward trajectory whose state-content mimics the reverse ordering of the original states. By Theorem 9.2 the operations constituting it are not the inverses of the original operations; by Principle 9.4 their enactment increments the count.

Remark 9.5 (Solution sets versus physical processes). The mathematical operation $T : x(t) \mapsto x(-t)$ is well defined as an automorphism of the solution set: it sends a solution to a solution, no more problematic than sending $f(x)$ to $f(-x)$ on a function space. But it does not correspond to a physical operation on a system; there is no laboratory procedure whose enactment maps a system from $x(t)$ to $x(-t)$. The mathematical operation acts on representations; physical operations act on systems. The conflation of these two levels is the source of Loschmidt’s paradox (Corollary 3.9).

10 Causal Distinguishability: An Independent Corollary

The preceding depths resolve the paradox without any appeal to observers. We record here a fourth, independent route to the same conclusion, useful because it needs nothing of the partition machinery—only the causal structure of spacetime.

Definition 10.1 (Causal partition). For an event E , the *causal partition* divides states into those causally downstream of E (with E in their past light cone) and those not. A state at spacetime point x is downstream of E at y iff y lies in the past light cone of x .

Theorem 10.2 (Causal distinguishability). *Let E_{rev} be the velocity-reversal event at time T , let S_0 be the state at the initial time $t_0 < T$, and let S'_0 be the post-reversal state at $2T$ whose configuration matches S_0 . Then E_{rev} is in the past light cone of S'_0 but not of S_0 ; the two states lie on opposite sides of the causal partition and are therefore distinct, independent of any observer.*

Proof. S_0 occurs before E_{rev} ($t_0 < T$), so E_{rev} is not in its past light cone: S_0 is not downstream of the reversal. S'_0 occurs after E_{rev} ($2T > T$) and the system reached it by evolving through E_{rev} , so E_{rev} is in its past light cone: S'_0 is downstream. Lying on opposite sides of the causal partition, $S_0 \neq S'_0$. $\square$

The matching configuration at $2T$ is not a return to t_0 ; it is a distinct state at a later time, carrying the reversal event in its causal past. This is the same conclusion as Theorem 6.1—the observable coincides while the history does not—reached now from spacetime geometry alone.

Remark 10.3 (Velocity through zero, and the double occupation). A continuous velocity reversal passes the velocity through $v = 0$ at the turning point, and the procedure is claimed to return the system to its initial configuration. A genuine return would make the initial state occur *twice*—once at the start, once on return at a strictly later count. By Theorem 3.6 the later occurrence carries a strictly greater count, so the two occurrences are distinct states with the same configuration, never the same state revisited. The nearest genuine recurrence of a configuration lies a full oscillatory period later (Proposition 3.4); it is a forward advance to a resembling configuration, not a return to the original moment.

11 Objections and Replies

“But the equations *are* time-reversible.” Granted, and irrelevant. Time-reversal invariance (2) is a property of the solution set (Remark 2.1). The claim of this paper is not that the equations lack the symmetry but that the symmetry is not an operation—there is no physical procedure whose enactment realises R on a system (Corollary 3.9, Remark 9.5). The mathematics is untouched; only its operational interpretation is corrected.

“The spin echo really does reverse the dynamics.” It does not. By Theorem 6.1 the echo recovers the *observable* by a forward operation on a *preserved* microstate while the *count* advances; nothing is run backward. The very part of the evolution that *is* a committed dynamical process—the homogeneous, partition-event part—is precisely the part the echo cannot recover (Theorem 7.3). The echo is the cleanest possible demonstration that “the signal returned” is not “the system went back.”

“In a block universe all moments coexist; there is no flow to reverse.” The account here is neutral on the metaphysics of becoming. Whether or not all moments “coexist,” the operational quantity M is monotone along every worldline (Theorem 3.6) and no operation decrements it (Theorem 3.7). The eternalist may read M as a monotone labelling of events along a worldline rather than a flowing now; the conclusion—no operation realises decrement—is unchanged.

“The continuum limit $f \rightarrow \infty$ erases the discreteness you rely on.” The continuum parameter t is the $f \rightarrow \infty$ idealisation of the count (Theorem 3.5); taking the limit does not make the count decrementable, it only makes its steps unresolvably fine. Monotonicity survives every finite f and its limit. The discreteness is a convenience of exposition, not a premise of the argument.

“This is just Boltzmann’s improbability restated.” No. Boltzmann grants that reversal is possible and explains why it is not seen (Table 1). The present claim is stronger and different in kind: reversal is not improbable but *incoherent as an operation* (Theorem 3.7), and this is witnessed, not merely argued, by the irreducible floor of the echo progression (Theorem 8.2).

12 Falsifiable Predictions

The account is empirical, not merely interpretive, and makes definite predictions.

1. **Engineering-independence of the floor.** In any many-body reversal experiment, as control imperfections are reduced, the residual fidelity-decay rate approaches a positive limit set by the system’s intrinsic dynamics and *independent* of the residual perturbation strength (Principle 8.1). A reversal protocol whose residual decay vanished with improving control would falsify the account.
2. **Floor- T_2 identity.** The Hahn-echo floor is $e^{-2\tau/T_2}$ with T_2 the homogeneous relaxation time measured independently; the count-deficit $1 - e^{-2\tau/T_2}$ must track the independently measured population of committed partition events (Corollary 7.4). A discrepancy between the echo deficit and the independently measured homogeneous relaxation would falsify the identification.
3. **Positivity of the smallest knowable error.** No bounded observer’s reconstruction of an initial state by engineered reversal can have strictly zero error; the minimum is set by the smallest committed correlation not pre-compensable from the present microstate (Theorem 7.3). A demonstrated zero-error reversal of a genuinely interacting many-body evolution would falsify the account.

13 Discussion

Reversible computing. Even logically reversible gates (Bennett, 1973), and the Landauer bound on erasure (Landauer, 1961), operate within the count: executing a reversible gate is a forward operation incrementing M (Theorem 3.8). Reversible computing minimises the μ -level entropy generated per logical step; it cannot make the global count decrement, and so cannot run time backward.

Measurement. A measurement commits which-system information to an apparatus—a partition event in the sense of Definition 7.1. The irreversibility of measurement is then not an extra postulate but an instance of Theorem 7.3: the committed correlation is not a function of the system’s present microstate alone and cannot be refocused away.

Quantum chaos. The Loschmidt-echo / fidelity-decay literature (Gorin et al., 2006; Jalabert and Pastawski, 2001) already measures, under another name, exactly the separating quantity of Theorem 8.2. The perturbation-independent decay regime is the floor; our account predicts its positivity and identifies it with the committed partition count.

The arrow of time. On this account the arrow is neither imposed by special initial conditions (Penrose, 1989) nor a matter of coarse-graining (Gibbs, 1902) nor of ignorance (Jaynes, 1957). It is constitutive: time *is* the monotone count (Theorem 3.5), and the count cannot decrease because counting is what time is (Theorem 3.7).

14 Conclusion

Loschmidt’s paradox asks how the time-reversal invariance of the microscopic equations can coexist with the irreversibility of the macroscopic world. We have argued that the question is malformed, and we have argued it at four depths, each independently sufficient and each sharper than the last.

Time is a count (Section 3): elapsed time is the monotone partition count M , no operation decrements it, and the mathematical time-reversal map acts on representations rather than systems—so “is physical dynamics reversible?” is a category error. *The paradox feeds on a conflation* (Section 4): it identifies the observable, the microstate, and the count, which are logically independent; once they are separated, a Poincaré recurrence is seen to return the microstate but never the count, dissolving the Zermelo objection. *The separation is witnessed in the laboratory* (Sections 5–8): the spin echo recovers the observable by a forward operation on a preserved microstate while the count advances, and the seventy-year progression of reversal experiments exposes an irreducible, engineering-independent floor that is the count made visible. *No reverse operation exists* (Section 9): physical trajectories are sequences of constitutively one-way operations, the reversibility of the flow description is a coarse-graining artifact, and every “rewind” is a forward operation incrementing the count.

The Second Law is, on this account, not a statistical regularity overlaid on reversible foundations but the operational signature of what time is. The completion of the resolution is therefore not a stronger argument but an empirical fact: the experiment named after Loschmidt has been refuting Loschmidt’s premise since 1950.

References

- Charles H. Bennett. Logical reversibility of computation. *IBM Journal of Research and Development*, 17(6):525–532, 1973.
- Felix Bloch. Nuclear induction. *Physical Review*, 70(7-8):460–474, 1946.
- Nicolaas Bloembergen, Edward M. Purcell, and Robert V. Pound. Relaxation effects in nuclear magnetic resonance absorption. *Physical Review*, 73(7):679–712, 1948.
- Ludwig Boltzmann. Über die beziehung zwischen dem zweiten hauptsatze der mechanischen wärmetheorie und der wahrscheinlichkeitsrechnung respektive den sätzen über das wärmegleichgewicht. *Sitzungsberichte der Akademie der Wissenschaften, Wien*, 76:373–435, 1877.
- Ludwig Boltzmann. Entgegnung auf die wärmetheoretischen betrachtungen des Hrn. E. Zermelo. *Annalen der Physik*, 293(4):773–784, 1896.
- Herman Y. Carr and Edward M. Purcell. Effects of diffusion on free precession in nuclear magnetic resonance experiments. *Physical Review*, 94(3):630–638, 1954.
- Arthur S. Eddington. *The Nature of the Physical World*. Cambridge University Press, Cambridge, 1928.

- Louis Essen and Jack V. L. Parry. An atomic standard of frequency and time interval: A caesium resonator. *Nature*, 176:280–282, 1955.
- Josiah Willard Gibbs. *Elementary Principles in Statistical Mechanics*. Charles Scribner’s Sons, New York, 1902.
- Thomas Gorin, Tomaž Prosen, Thomas H. Seligman, and Marko Žnidarič. Dynamics of loschmidt echoes and fidelity decay. *Physics Reports*, 435(2-5):33–156, 2006.
- Erwin L. Hahn. Spin echoes. *Physical Review*, 80(4):580–594, 1950.
- Rodolfo A. Jalabert and Horacio M. Pastawski. Environment-independent decoherence rate in classically chaotic systems. *Physical Review Letters*, 86(12):2490–2493, 2001.
- Edwin T. Jaynes. Information theory and statistical mechanics. *Physical Review*, 106(4):620–630, 1957.
- Rolf Landauer. Irreversibility and heat generation in the computing process. *IBM Journal of Research and Development*, 5(3):183–191, 1961.
- Patricia R. Levstein, Gonzalo Usaj, and Horacio M. Pastawski. Attenuation of polarization echoes in nuclear magnetic resonance: A study of the emergence of dynamical irreversibility in many-body quantum systems. *The Journal of Chemical Physics*, 108(7):2718–2724, 1998.
- Joseph Liouville. Note sur la théorie de la variation des constantes arbitraires. *Journal de Mathématiques Pures et Appliquées*, 3:342–349, 1838.
- Josef Loschmidt. Über den zustand des wärmeleichgewichtes eines systems von körpern mit rücksicht auf die schwerkraft. *Sitzungsberichte der Akademie der Wissenschaften, Wien, Mathematisch-Naturwissenschaftliche Klasse*, 73:128–142, 1876.
- Josef Loschmidt. Über den zustand des wärmeleichgewichtes eines systems von körpern (fortsetzung). *Sitzungsberichte der Akademie der Wissenschaften, Wien*, 75:67–73, 1877.
- Saul Meiboom and David Gill. Modified spin-echo method for measuring nuclear relaxation times. *Review of Scientific Instruments*, 29(8):688–691, 1958.
- Roger Penrose. *The Emperor’s New Mind: Concerning Computers, Minds, and the Laws of Physics*. Oxford University Press, Oxford, 1989.
- Roger Penrose. *The Road to Reality: A Complete Guide to the Laws of the Universe*. Jonathan Cape, London, 2004.
- Henri Poincaré. Sur le problème des trois corps et les équations de la dynamique. *Acta Mathematica*, 13:1–270, 1890.
- Ilya Prigogine. *From Being to Becoming: Time and Complexity in the Physical Sciences*. W. H. Freeman, San Francisco, 1980.
- Won-Kyu Rhim, Alexander Pines, and John S. Waugh. Time-reversal experiments in dipolar-coupled spin systems. *Physical Review B*, 3(3):684–696, 1971.
- Ernst Zermelo. Über einen satz der dynamik und die mechanische wärmetheorie. *Annalen der Physik*, 293(3):485–494, 1896.
- Wojciech H. Zurek. Decoherence, einselection, and the quantum origins of the classical. *Reviews of Modern Physics*, 75(3):715–775, 2003.